

Lenskart - E-commerce Database Management System

Chapter 2

2.1 Introduction

The entities required for the Lenskart E-Commerce Database are identified based on eyewear shopping requirements, including eyeglasses, sunglasses, contact lenses, and prescription-based ordering. The attributes of each entity along with their Primary Keys (PK) and Foreign Keys (FK) are defined to maintain proper data organization. The relationships between entities are analysed using cardinality and participation constraints. This design helps in creating the ER diagram, relational schema, and database implementation while ensuring data consistency and reducing redundancy.

2.2 Entities, Attributes and Keys

Entity: Customer (Strong Entity)

Attribute	Key Type
Customer_ID	Primary Key
Customer Name	Not Null
Email	Unique Key
Phone No	Unique Key
Date of Birth	-
Address	-

Entity: Category (Strong Entity)

Attribute	Key Type
Category_ID	Primary Key
Category Name	Unique Key
Description	-

Entity: Brand (Strong Entity)

Attribute	Key Type
Brand_ID	Primary Key
Brand Name	Not Null
Contact_No	Unique Key
Address	-

Entity: Product (Strong Entity)

Attribute	Key Type
Product_ID	Primary Key
Product Name	-
Category_ID	Foreign Key
Brand_ID	Foreign Key
Frame_Type	-
Price	-

Stock	-
Power_Compatible	-

Entity: Prescription (Strong Entity)

Attribute	Key Type
Prescription_ID	Primary Key
Customer_ID	Foreign Key
Right_Eye_Power	-
Left_Eye_Power	-
Cylindrical_Power	-
Axis	-
Prescription_Date	-

Entity: Cart (Strong Entity)

Attribute	Key Type
Cart_ID	Primary Key
Customer_ID	Foreign Key
Total_items	-
Total_amount	-

Entity: Order (Strong Entity)

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_date	-
Order_status	Check Key
Total_amount	-

Entity: Order_Item (Weak Entity)

Attribute	Key Type
Order_item_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Prescription_ID	Foreign Key
Quantity	-
Price	-

Entity: Payment (Strong Entity)

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_method	-
Payment_status	Check Key

Payment_date	-
Amount	-

Entity: Review (Strong Entity)

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	Check Key
Review_text	-
Review_date	-

Entity: Delivery (Strong Entity)

Attribute	Key Type
Delivery_ID	Primary Key
Order_ID	Foreign Key
Delivery_address	-
Delivery_date	-
Delivery_status	Check Key

Entity: Store (Strong Entity)

Attribute	Key Type
Store_ID	Primary Key
Store_Name	-
Location	-
Contact_No	Unique Key

2.3 Entity Relationships

Entity 1	Relation	Entity 2	Cardinality	Participation
Customer	Owns	Cart	One-to-One	Partial/Total
Customer	Places	Order	One-to-Many	Partial/Total
Customer	Makes	Payment	One-to-Many	Partial/Total
Customer	Writes	Review	One-to-Many	Partial/Total
Customer	Has	Prescription	One-to-Many	Partial/Total
Category	Categorizes	Product	One-to-Many	Total/Total
Brand	Supplies	Product	One-to-Many	Partial/Total
Product	Included-in	Order_Item	One-to-Many	Partial/Total
Prescription	Applied-to	Order_Item	One-to-Many	Partial/Total
Order	Contains	Order_Item	One-to-Many	Total/Total
Order	Generates	Payment	One-to-One	Total/Total
Order	Generates	Delivery	One-to-One	Total/Total
Product	Receives	Review	One-to-Many	Partial/Total